

■ AI-Based Emotion Recognition Playlist Maker

Project Summary Report

This project is an AI-powered music recommendation web application that detects user emotions using text input and facial expressions, then recommends mood-based playlists using local recommendation logic.

Project Overview

The application combines: Emotion recognition Natural language processing Facial analysis Music recommendation logic Emotion-based playlist recommendation Responsive frontend UI

Live Demo: <https://music-1-8f4y.onrender.com>

Core Features

1. Text Emotion Detection

- NLP-based emotion analysis
- Keyword matching using text2emotion
- Supports English and Hindi language inputs

2. Facial Emotion Detection

- Webcam capture using OpenCV
- Emotion recognition using DeepFace
- Confidence score generation

3. Combined Emotion Analysis

- Text-only analysis
- Face-only analysis
- Combined confidence-based selection

4. Playlist Recommendation System

- Emotion-to-mood mapping
- Local playlist recommendation logic
- Offline recommendation support

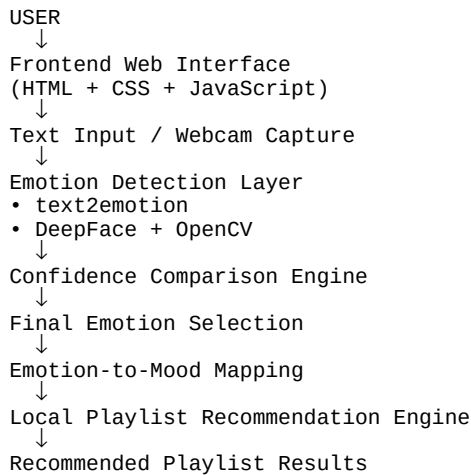
5. Responsive User Interface

- Animated gradient UI
- Mobile responsiveness
- Interactive frontend experience

Technology Stack

Category	Technologies Used
Frontend	HTML5, CSS3, JavaScript, Font Awesome
Backend	Python, Flask
AI/ML	DeepFace, text2emotion, TensorFlow, OpenCV
Data Processing	NumPy, Regex
Recommendation	JSON-based Playlist Mapping
Deployment	Render Cloud Hosting

System Architecture Diagram



Detailed Workflow

1. User opens the application.
2. User enters text or captures facial expression.
3. Frontend sends data to Flask backend.
4. Backend performs text emotion analysis.
5. DeepFace processes facial expressions.
6. Confidence scores are compared.
7. Best emotion is selected.
8. Emotion mapped to recommendation category.
9. Playlist recommendations generated.
10. Results displayed on frontend.

Project Structure

```
music-recommendation/
├──
├── app.py
├── requirements.txt
├── README.md
├── templates/
├── index.html
├── static/
├── css/
└── js/
```

Important Dependencies

```
Flask==2.3.3
text2emotion==0.0.5
opencv-python==4.8.0.76
deepface==0.0.79
tf-keras==2.15.0
numpy<2.0.0
requests==2.31.0
```

Future Improvements

• Voice emotion recognition • Real-time video emotion tracking • Personalized recommendations • Emotion analytics dashboard • Database integration • Mobile application support • AI chatbot integration

Conclusion

This project demonstrates the integration of Artificial Intelligence, Natural Language Processing, Computer Vision, and Full-Stack Web Development to create an intelligent emotion-based playlist recommendation system. The project successfully combines: Text emotion analysis Facial emotion recognition Recommendation logic Frontend and backend integration The project is suitable for: Academic major projects AI/ML portfolio showcases Internship

demonstrations Emotion-aware recommendation systems

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